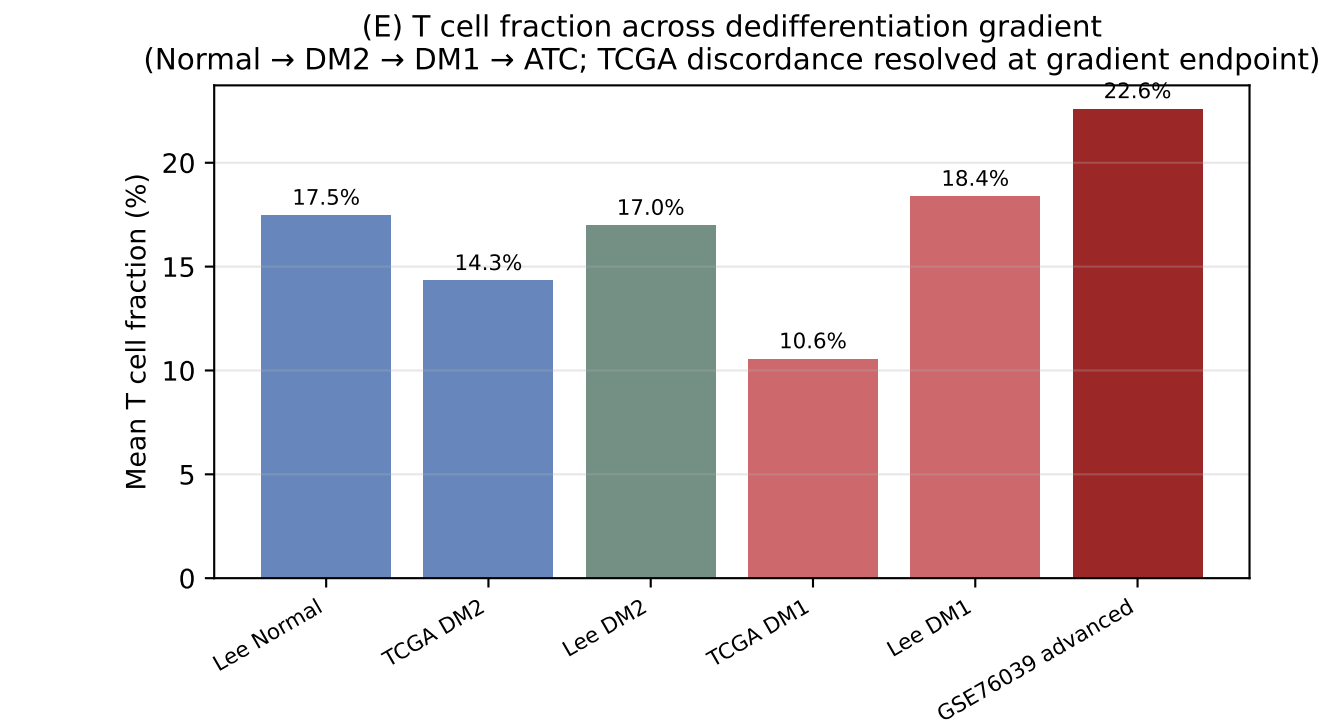
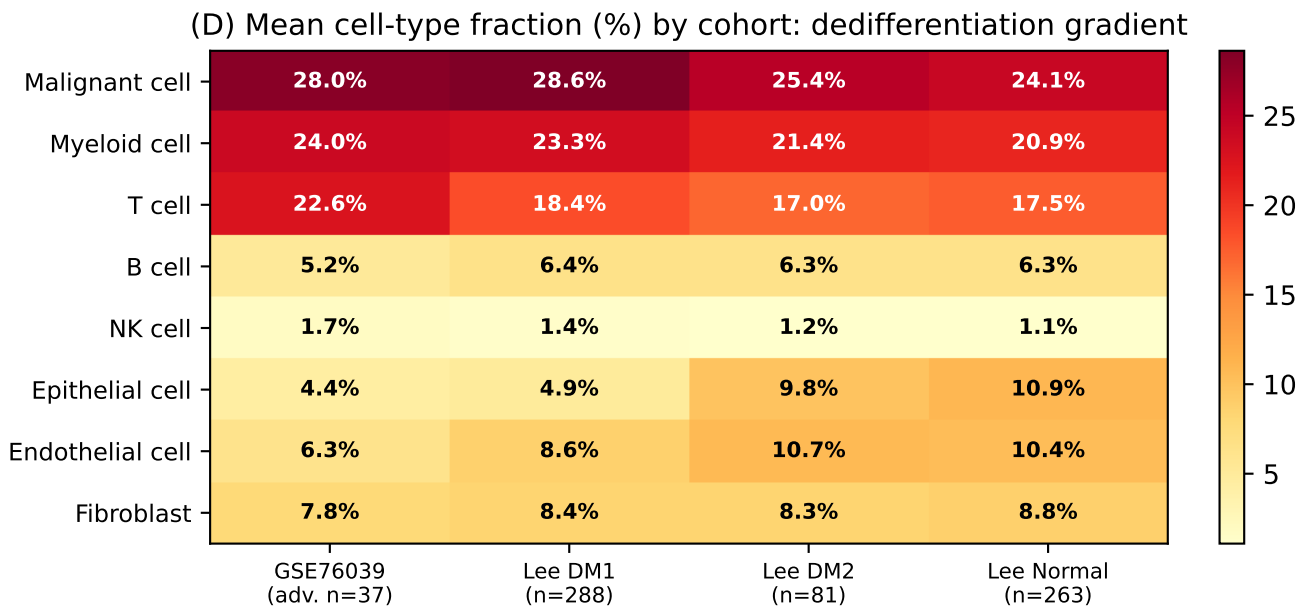
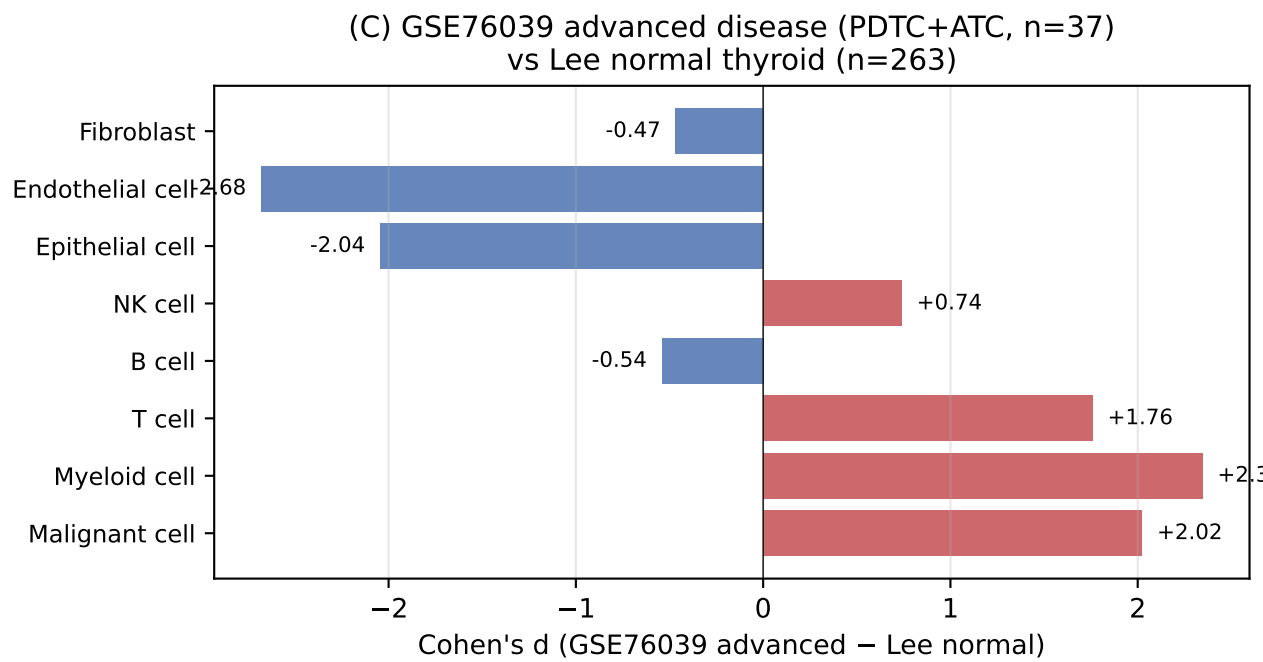
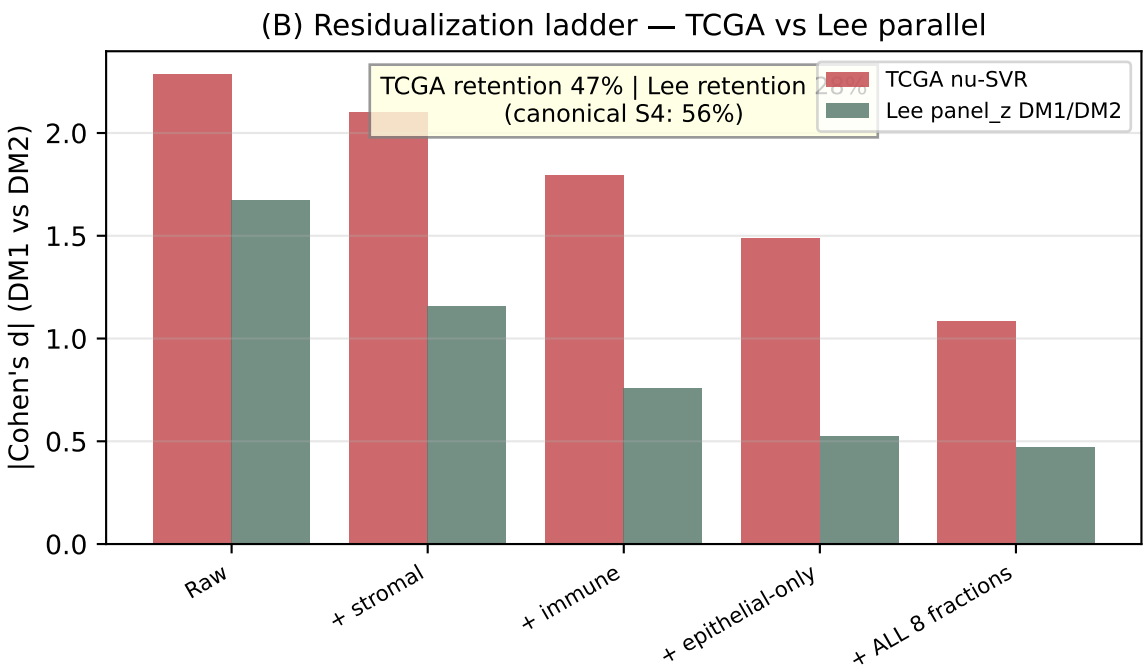
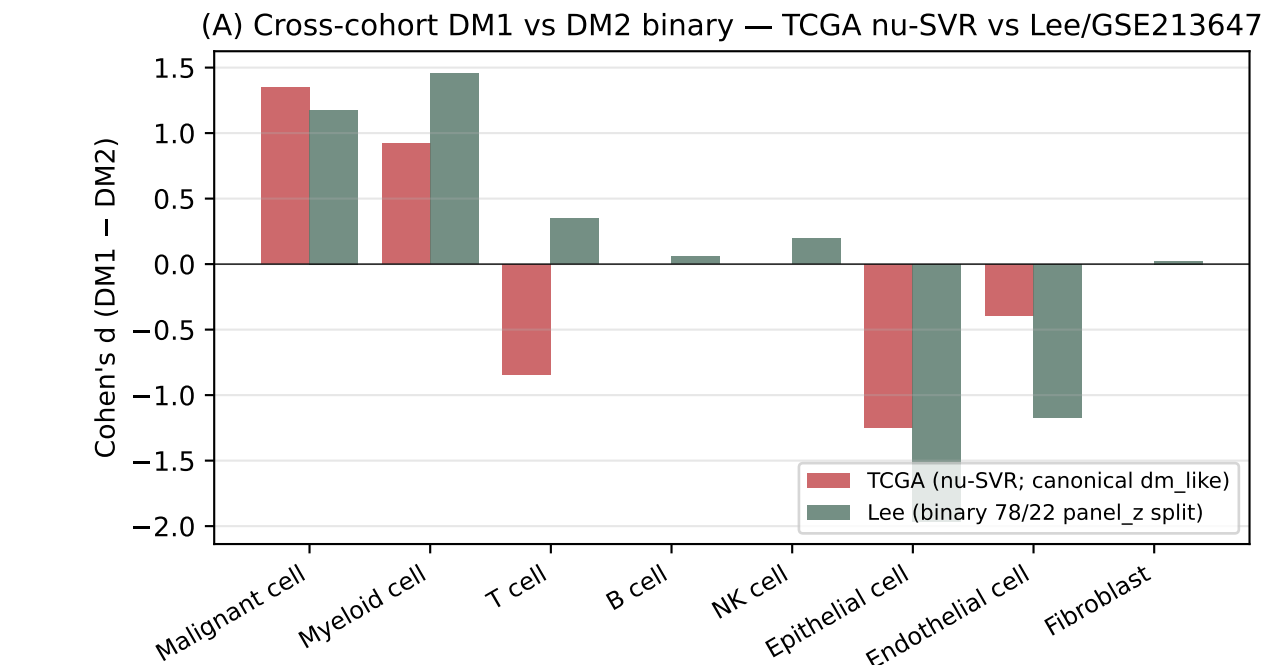




v4 EXTENSIONS — KEY FINDINGS

- (A) TCGA × Lee DM1 vs DM2 binary direction PERFECT MATCH
Both: Malignant↑, Myeloid↑ in DM1; Epithelial↓, Endothelial↓ in DM1
Effect sizes within 30% of each other across cohorts
- (B) RESIDUALIZATION LADDER PARALLEL
TCGA nu-SVR: 47% retained (matches S4 canonical 56%)
Lee panel_z DM1/DM2: 28% retained
Both retain non-zero signal after full residualization → cohort-level differentiation signature is real, not pure composition artifact
- (C+D) GSE76039 ADVANCED DISEASE EXTENDS GRADIENT
PDTC+ATC end of dedifferentiation:
- Endothelial d=-2.68 (massive vascular loss)
- Myeloid d=+2.35 (massive infiltration)
- Epithelial d=-2.04 (loss of normal thyrocytes)
- Malignant d=+2.02 (tumor cell dominance)
- T cell d=+1.76 (advanced inflammation)
- (E) T CELL DISCORDANCE RESOLVED
Apparent TCGA DM1 < DM2 T cell fraction was binary-collapse artifact.
Across the full Lee Normal → DM2 → DM1 → GSE76039 gradient:
8.4% → 17.0% → 18.4% → 22.6% — monotonic INCREASE.
DM1 ≈ "advanced-disease direction" — Q10 reviewer-defense extended.

CROSS-COHORT VERDICT (n=572 TCGA + 632 Lee + 37 GSE76039 = 1,241 bulk samples)
Direction-consistent for Malignant / Myeloid / Epithelial / Endothelial in DM1-direction across all cohorts. T cell shows progressive enrichment along the dedifferentiation gradient (binary TCGA snapshot was misleading).

Bulk-level RAI score residualization on cell-type composition retains 28-47% of effect across cohorts; the patient-level resolution remains the Pu 2021 sc per-patient analysis (Paper 1 Figure 3).

CAVEATS

- scaden 5th method retry FAILED at process step (sklearn MinMaxScaler shape-mismatch on simulated h5ad samples; library bug, not data bug). Defer to BayesPrism R port post-bioRxiv.
- GSE76039 microarray (22,880 genes) intersect with Lu HVG only 1,221 genes (61%) — fewer than RNA-seq cohorts. Direction findings robust; absolute fractions less precise.
- MSK-IMPACT cohort is targeted DNA panel (no RNA-seq) — cross-cohort bulk RNA validation limited to TCGA + Lee + GSE76039.